DIS - Data Extraction Strategy

Criteria	App 1: Doc AI + Gemini OCR	App 2: Doc AI + Vertex AI (AutoML)	App 3: Doc AI (Classification) + Vertex AutoML (Extraction)	App 4: Doc AI (Classification) + Vertex Custom Training
Architecture	OCR + Generative AI (prompt-based extraction)	OCR + AutoML trained model	Doc AI classification + AutoML extraction	Doc AI classification + custom ML model
Pros	• No training required for gemini and less data for Doc AI - Fast implementation - Works with very small datasets - Flexible for unstructured docs	• Structured ML-based extraction - Lower coding effort than custom ML - Managed training pipeline	• Separation of concerns (classification + extraction) - Better control than GenAI	• Full control over model design - Highest potential accuracy
Cons	• Output variability (non-deterministic) - Prompt	• Requires labeled dataset - Limited flexibility	• Increased pipeline complexity - Still requires	• High development effort - Requires ML expertise

	engineering required	vs custom ML	labeled data	and MLOps
Ease of Development	Very High	High	Medium	Low
Time to Build	1–2 weeks	2-3 weeks	3-4weeks	4–8+ weeks
Data Requirement	Very Low	Medium	Medium	High
Accuracy (Low Data Scenario)	High (due to pre-trained LLMs)	Moderate	Moderate	Low (insufficient training data)
Scalability	Medium-High	High	High	Very High
Maintenance Effort	Low	Medium	Medium–High	High
Best Fit Use Case	POC, MVP, low-data environments	Structured document processing	Modular enterprise pipelines	Large-scale enterprise with high data availability
Accuracy (Low Data Scenario)	High (due to pre-trained LLMs)	Moderate	Moderate	Low (insufficient training data